Tributary

The experiment data platform for drug discovery, from raw files to decisions.

Proteomics-first · Omics meets chemistry · Installs in your own AWS account
Open formats · Audit trail and SSO included · AI assistants with governed tools

Product overview · 2026

THE PROBLEM

Discovery data is scattered, and decisions wait for it

Drug-discovery teams generate proteomics, transcriptomics and screening data every week, but it lives in instrument folders, spreadsheets and one-off scripts. Comparing experiments, linking results to compounds, or simply finding last quarter's run takes days of a scientist's time.

57%
of life-science lab experts name **data silos** as their top challenge.

49%
say they most need help with **data standards and ontologies** to make data FAIR.

Source: Pistoia Alliance, Lab of the Future 2025 survey (200+ experts), reported by IPT Online.

What teams tell us they lack

- One place where an experiment's samples, files, processing and results stay together, versioned.
- Proteomics treated as a first-class data type, not squeezed into NGS-shaped tools.
- Results that meet chemistry: "what does compound X do across all experiments?"
- A platform small biotechs can afford, with SSO and an audit trail that are not enterprise add-ons.
- AI help they can trust: answers that show their query and respect permissions.

THE PRODUCT

One platform, from sample sheet to decision

Tributary registers experiments with structured, ontology-backed metadata, runs standard pipelines, imports tool outputs into versioned datasets, and lets scientists explore, join, chart and decide across experiments, all inside the customer's own cloud account.

Registry

Experiments, versions and sample manifests validated against templates and ontologies. Published versions are immutable; every change needs a reason and is audited. SDRF import and export.

Pipelines and ingest

Launch the Nextflow pipelines your admins register (e.g. quantms) on AWS Batch or HealthOmics. Large files upload straight to object storage; QC gates before import; bulk onboarding of legacy folders.

Data catalogue

Importers for DIA-NN, Spectronaut, MaxQuant, FragPipe, AlphaDIA, quantms, MSstats, limma, DEqMS, DESeq2, MAGeCK and CurveCurator. Versioned releases with data contracts, quality rules and lineage.

Explore and chart

Fast queries over Parquet with DuckDB; joins to samples, compounds and annotations; 16 chart types including volcano, clustered heatmap, PCA and dose-response. Statistics run in a separate, tested service.

Decide

Protein and compound pages, contrast comparison, pathway enrichment, a hit triage board with reasons and history, watchlists with alerts, cross-experiment QC.

Chemistry and structure

Compound structures and descriptors, substructure and similarity search (RDKit); 3D protein models coloured by confidence (Mol*); gene–concept networks.

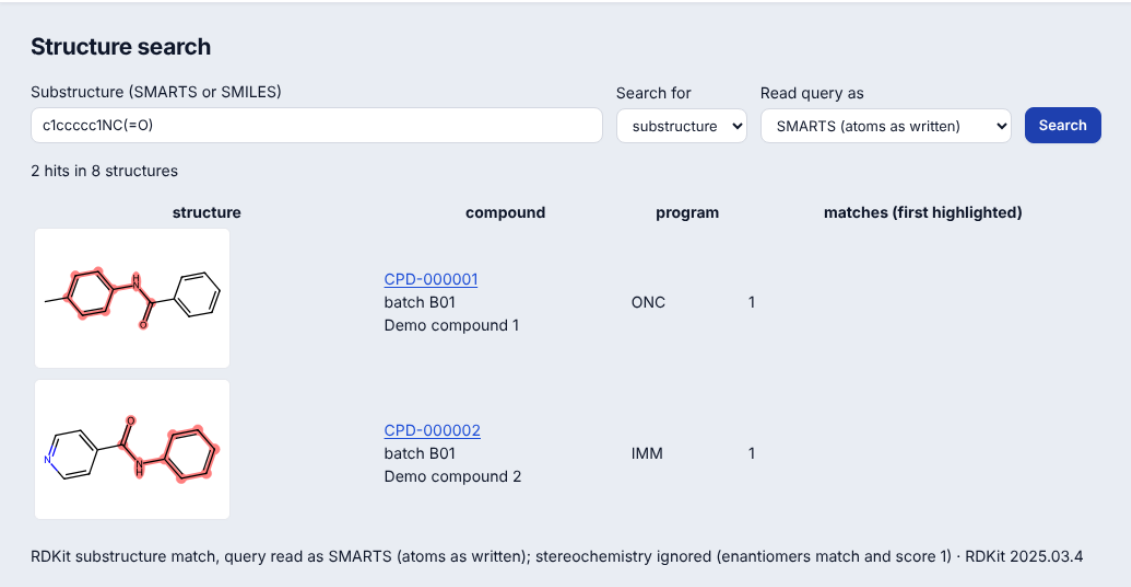
Every chart has a table alternative; every number on screen links back to the release, version and file it came from.

IN THE PRODUCT

What scientists see



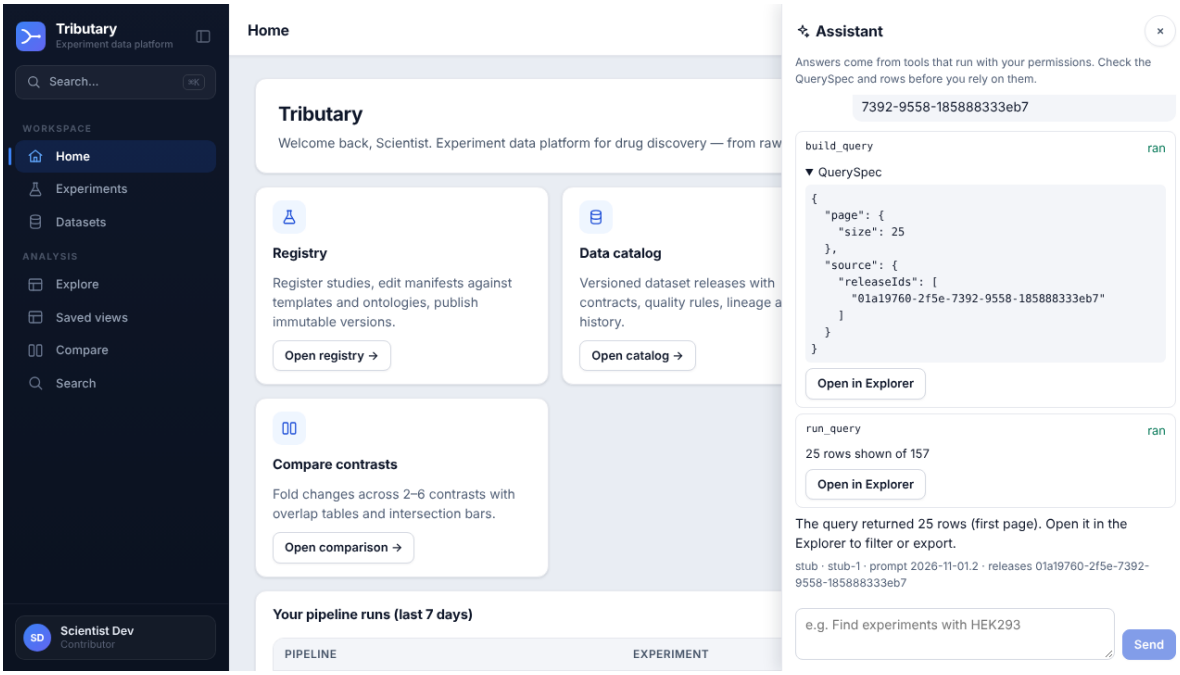
Cross-experiment QC: identifications per run across experiments, low-quality runs flagged, and a PCA of runs coloured by plate or date to expose batch effects.



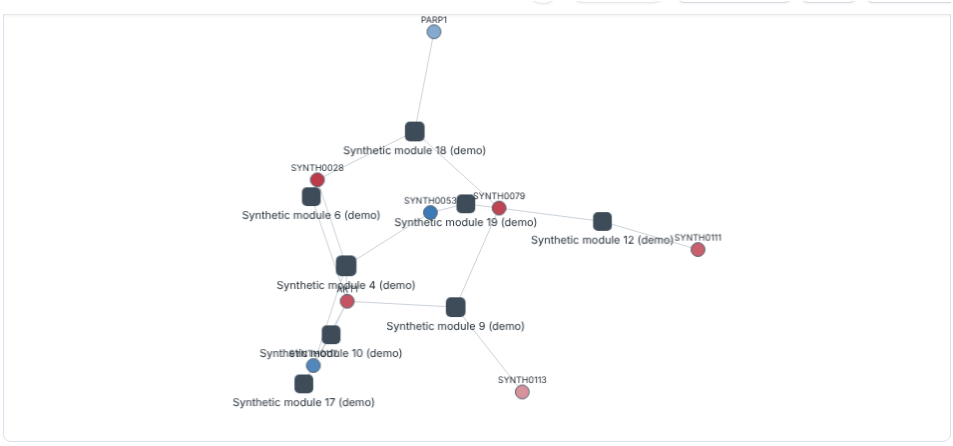
Structure search: substructure (SMARTS or SMILES) and similarity search over the compound registry, with the match highlighted.

IN THE PRODUCT

Answers you can check



The assistant answers by calling governed, deterministic tools with the user's own permissions, and shows exactly the query it ran.



Gene–concept network of an enrichment result: which hit genes drive which pathways, coloured by fold change, with a table alternative.

WHY TRIBUTARY

Where we are different

	What you get
Proteomics	First-class: native importers for the main DIA and DDA tools, dose-response proteomics, QC
Omics + chemistry	Joins results to compound IDs, ADME properties, targets and structures
Where data lives	In the customer's own AWS account and region; customer-managed keys
Formats	Open: Parquet/Iceberg, SDRF, a documented REST API (OpenAPI), a Python SDK and MCP for AI tools. No lock-in
Trust	Immutable versions, reasons for change, full audit log, lineage on every result
AI	Assistants call validated tools, never free-form SQL; writes need human approval
Cost to run	Estimated USD 430–520 per month of infrastructure per environment, before compute

Infrastructure estimate from the reference AWS architecture (single environment, x86); actual cost depends on region, data volume and usage.

Who it is for

Bench scientists
Register an experiment from a sample sheet, launch the standard pipeline, and see QC, volcano and dose-response results the same day.

Computational biologists
Query, filter and join across experiments; export to notebooks with copy-as-code; trust lineage instead of rebuilding it.

Programme leads and chemists
"What does compound X do across all experiments?" Compare compounds on targets and ADME; triage hits with reasons.

Data and platform admins
Templates, picklists, ontologies, pipelines, roles and audit in one place, in an environment the company controls.

SECURITY AND DEPLOYMENT

Built for regulated science

Identity and access
Single sign-on through OIDC (Okta, Entra ID, Auth0, Keycloak). Per-program roles enforced in the API; group-to-role mapping; scoped, revocable API tokens.

Data stays with you
Runs in your AWS account and region. Encryption at rest with customer-managed KMS keys, TLS everywhere. Large files never pass through the application.

Audit and integrity
Append-only audit log of every change, export and AI tool call; reasons for change; re-authentication on publish; immutable published data.

Part 11-ready controls
The controls regulated work needs are built in. Tributary is not a validated GxP system; validation remains the customer's process.

Software supply chain
Pinned dependencies, SBOM and dependency review in CI, licence allowlist checked across every package; no GPL code in the product.

Deployment options
Terraform modules for AWS (production); single-container trial for a laptop or small cloud VM; documented runbook, health checks and upgrades.

How a pilot works

- **Week 1**: deploy into your AWS account (or a trial VM), connect SSO, import a few past experiments.
- **Weeks 2–4**: your scientists register new experiments, run the standard pipeline and use results pages, QC and triage on real programmes.
- **Week 5**: review together: time to results, data found, what to configure next.

Coming next

- Protein–protein interaction networks (STRING) around hits.
- Connectors to electronic lab notebooks and compound registries.
- One-click experiment reports for programme meetings.
- Email and chat notifications for alerts; PTM site-level data.

Start a pilot
Bring one programme and a handful of past experiments. We will have them searchable, comparable and charted in your own environment within the first week. Contact: [your sales contact].

Tributary is a working name. All screenshots show synthetic demonstration data. Product names of third-party tools belong to their owners.